Sedimentation Rate, Modified Westergren

Test Details

METHODOLOGY

Modified Westergren; automated and manual methods are employed.

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Evaluate the nonspecific activity of infections, inflammatory states, autoimmune disorders, and plasma cell dyscrasias

Limitations

Optimum results are from blood less than two hours old. The ESR is of limited diagnostic value in severe anemia or in hematologic states that affect increased size and shape variation (poikilocytosis) of the RBC (ie, presence of sickle cells or spherocytes). Extreme plasma viscosity will result in a decreased ESR.

Custom Additional Information

Elevations in fibrinogen, α- and β-globulins (acute phase reactants), and immunoglobulins increase the sedimentation rate of red cells through plasma. The test is important in the diagnosis of temporal arteritis, as well as its management.[1]

Specimen Requirements

Specimen

Whole blood

Volume

Tube fill capacity

Minimum Volume

2 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Lavender-top (EDTA) tube

Collection Instructions

Invert tube immediately 8 to 10 times once tube is filled at time of collection.

Stability Requirements

Temperature	Period
Refrigerated	24 hours

Reference Range

- Male: 0 to 50 years: 0−15 mm/hour, 50 years and older: 0−30 mm/hour

- Female: 0 to 50 years: 0−32 mm/hour, 50 years and older: 0−40 mm/hour

Storage Instructions

Refrigerate

Causes for Rejection

Hemolysis; clotted specimen; underfilled tube; specimen older than 24 hours; improper labeling; transfer tubes with whole blood; specimen received in any anticoagulant other than EDTA; specimen diluted or contaminated with IV fluid; patient specimen with presence of cold agglutinins or cryoglobulins; specimen received with plasma removed

References

Gambino SR, Dire JJ, Monteleone M, et al. The Westergren sedimentation rate using K_3 EDTA. *Tech Bull Regist Med Technol.* 1965; 35:1-8.

Harmening D. *Clinical Hematology and Fundamentals of Hemostasis.* 2nd ed. Philadelphia, Pa: Lippincott;1992:532-534.

Footnotes

1. Wong RL, Korn JH. Temporal arteritis without an elevated erythrocyte sedimentation rate. Case report and review of the literature. *Am J Med.* 1986; 80(5):959-964. PubMed 3518441 (http://www.ncbi.nlm.nih.gov/pubmed/3518441)